

CET 251 Artificial Intelligence — Project Proposal

Project D4: Smart Heritage Tourism Recommender (SHTR)

Project ID & Title:	D4 - Smart Heritage Tourism Recommender
Team Members:	Member 1 (AI Lead), Member 2 (App Lead), Member 3 (QA Lead), Member 4 (Doc Lead)
Problem Statement:	Visitors to Cairo & Giza face information overload and struggle to find heritage attractions matching their strict budget, available time, and accessibility needs.
Target Users:	Cultural tourists, budget-conscious travelers, students, and visitors requiring wheelchair access.
Project Objective:	Build a functional, tested AI prototype recommending tailored tourism sites using hard constraint filtering and TF-IDF similarity matching.
In-Scope (MVP):	40 curated sites in Cairo & Giza, hard filters (budget, duration, location, wheelchair, opening period), TF-IDF Cosine Similarity recommendations, Popularity Baseline fallback, Streamlit web app.
Out-of-Scope:	City-scale global routing, online payment processing, live GPS tracking.
AI Formulation:	Content-based filtering using TF-IDF vectorization and Cosine Similarity scoring against user interest tags, compared against a Popularity Baseline.
Dataset Plan:	40 heritage sites with 10 attribute fields stored in data/raw/egypt_attractions.csv.
Evaluation Plan:	Filter Match Rate metric (Target 100%) and 8 comprehensive test cases in tests/test_report.md.
Deployment Plan:	Local Streamlit web app running offline with 1-click launcher scripts (run.py, run.bat).
Instructor Decision:	☒ Approved ☐ Approved with conditions ☐ Revise and resubmit